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Discond-VAE: Disentangling Continuous Factors from the Discrete

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Abstract

We propose a variant of VAE capable of disentangling both variations within each class and variations shared across all classes. To represent these generative factors of data, we introduce two sets of continuous latent variables, *private variable* and *public variable*. Our proposed framework models the private variable as a Mixture of Gaussian and the public variable as a Gaussian, respectively. Each mode of the private variable is responsible for a class of the discrete variable.

Most of the previous attempts to integrate the discrete generative factors to disentanglement assume statistical independence between the continuous and discrete variables. However, this assumption does not hold in general. Our proposed model, which we call Discond-VAE, DISentangles the class-dependent CONtinuous factors from the Discrete factors by introducing the private variables. The experiments show that Discond-VAE can discover the private and public factors from data qualitatively and quantitatively.

1 Introduction

Learning disentangled representation of data without supervision has been considered an important task for representation learning. (Bengio, Courville, and Vincent 2013) Although there are diverse quantitative measures for the disentangled representation (Higgins et al. 2016) (Kim and Mnih 2018) (Chen et al. 2018) (Ridgeway and Mozer 2018) (Eastwood and Williams 2018) (Kumar, Sattigeri, and Balakrishnan 2017), most of the qualitative interpretation agrees on the statistical independence between each basic element of representation. In other words, each element of the disentangled representation corresponds to only one generative factor of data while being invariant to the others. Hence, the disentangled representation is naturally a concise and explainable feature of data. Recently, various VAE-based models have been proposed to obtain more disentangled representation such as (Higgins et al. 2016) (Kim and Mnih 2018) (Chen et al. 2018) (Eastwood and Williams 2018) (Kumar, Sattigeri, and Balakrishnan 2017) (Esmaeili et al. 2019) (Dupont 2018).

In particular, JointVAE (Dupont 2018) introduced discrete latent variables as well as continuous variables to represent the generative factors of data. For the real-world data, there are intrinsically discrete generative factors such as digit-type in the MNIST dataset. Therefore, it is natural to adopt a discrete variable to get a disentangled representation of those generative factors. However, JointVAE has a limitation of assuming the independence between the continuous and discrete variables.

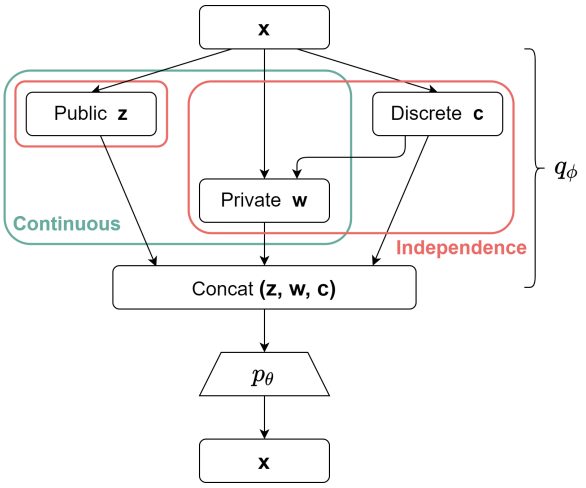

Figure 1: Overview of Discond-VAE. Discond-VAE introduces two continuous latent variables (public and private variables) and one discrete variable to represent the data $\mathbf{x}$. The public continuous latent variable is assume to be independent to the private and discrete variables.

The independence assumption between the continuous and discrete variables of JointVAE is too restrictive to the general data. For example, consider the CelebA dataset (Liu et al. 2015). The CelebA dataset has 40 attribute labels, including Male(Gender) and Mustache. In this case, a continuous generative factor representing the Mustache volume is not independent with a discrete factor of the Gender. Hence, the class-independent continuous variable of JointVAE cannot properly represent the Mustache volume. From a similar perspective, (Mathieu et al. 2019) proposed a generalization of the disentanglement, *Decomposition*. (Mathieu et al. 2019) defines the decomposition of the latent variable as imposing the desired structure to the aggregate posterior $q_\phi(\mathbf{z})$,

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such as sparsity or clustering, generalizing independence for disentanglement. (Mathieu et al. 2019) achieved a better disentangled by imposing an anisotropic prior to the β-VAE.

In this paper, we propose a new VAE model called Discond-VAE. Instead of imposing independence between the continuous and discrete factors of data as in the previous works, we propose learning the independent and dependent continuous factors jointly. Discond-VAE classifies the continuous latent variable into two groups, *private* and *public* variable. First, we refer to each category of the discrete generative factor of data as *class*. The *private* variable represents variation within each class and the *public* variable encodes the common generative factor of the entire classes. Therefore, Discond-VAE is able to represent the intra-class variation while keeping the capacity to represent the class-independent generative factor as in JointVAE.

Following the intuitive interpretation, we assume the public variable is independent with the discrete and private variables. The public and private variables are modeled by the Gaussian distribution and the Mixture of Gaussian, respectively. Each mode of the private variables corresponds to a class of the discrete variable. The experiments demonstrate that Discond-VAE can extract the private and public variables from data qualitatively and quantitatively.

1.1 Contribution

- We propose a new VAE model called Discond-VAE. To the best of our knowledge, Discond-VAE is the first VAE model to represent the public and private continuous generative factors and the discrete generative factors at the same time.
- We propose a CondSprites dataset reassembled from the dSprites (Matthey et al. 2017) to evaluate the representing ability for the private and public variables. The CondSprites dataset is designed to mimic the class-independent and the class-dependent generative factors of the real-world datasets.
- The existing disentanglement metrics assume the continuity of the latent variables and the independence of the generative factors. To integrate the discrete latent variable and class-dependent continuous variable to the disentanglement metrics, we propose a conditional disentanglement evaluation.
- We assess Discond-VAE on the MNIST, dSprites, and CondSprites. The experiments show that Discond-VAE can disentangle the public and private factors qualitatively and quantitatively.

2 Background

2.1 VAE

Variational Autoencoder (VAE, (Kingma and Welling 2013) (Rezende, Mohamed, and Wierstra 2014)) is a probabilistic model that learns a joint distribution $p(x, z)$ of the observed data x and a continuous latent variable $z \in \mathbb{R}^d$. VAE models the joint distribution by

$$p(\mathbf{z}) = \mathcal{N}(0, I_{d\times d}) \tag{1}$$

$$p_\theta(\mathbf{x} \mid \mathbf{z}) = p(\mathbf{x}; \mu_\theta(\mathbf{z})) \tag{2}$$

where $\mathcal{N}(0, I_{d\times d})$ denotes a d-dimensional isotropic Gaussian distribution and $p(\mathbf{x}; \mu_\theta(\mathbf{z}))$ is a probabilistic distribution model with distribution parameters $\mu_\theta(\mathbf{z})$ (e.g., Gaussian or Bernoulli). $p(\mathbf{x}; \mu_\theta(\mathbf{z}))$ is often referred to as the decoder.

Given a dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$, VAE model is optimized by the Maximum Likelihood Estimate framework (MLE). Since the exact inference of log-likelihood $\log p(\mathbf{x})$ requires a posterior distribution $p(\mathbf{z}|\mathbf{x})$ which is intractable, VAE applies the variational inference by introducing an inference network $q_\phi(\mathbf{z}|\mathbf{x})$, which is often referred to as the encoder. The encoder $q_\phi(\mathbf{z}|\mathbf{x})$ approximates the true posterior $p(\mathbf{z}|\mathbf{x})$ with a factorized Gaussian distribution with parameters encoded by the neural network.

$$q_\phi(\mathbf{z} \mid \mathbf{x}) = \prod_{j=1}^{d} \mathcal{N}(\mathbf{z}_j; \mu_j(\mathbf{x}), \sigma_j^2(\mathbf{x})) \tag{3}$$

The encoder $q_\phi(\mathbf{z}|\mathbf{x})$ and the decoder $p(\mathbf{x}; \mu_\theta(\mathbf{z}))$ is simultaneously optimized by maximizing the Evidence Lower Bound (ELBO) $\mathcal{L}_{\text{VAE}}(\theta, \phi)$.

$$\begin{aligned}\mathcal{L}_{\text{VAE}}(\theta, \phi) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x} \mid \mathbf{z})\right] \\ &\qquad - D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \mid\mid p(\mathbf{z})) \\ &= \log p(\mathbf{x}) - D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \mid\mid p_\theta(\mathbf{z} \mid \mathbf{x})) \\ &\geq \log p(\mathbf{x})\end{aligned} \tag{4}$$

The first term of the ELBO is the reconstruction loss which encourages the VAE to encode informative latent variables $\mathbf{z}$ to reconstruct the data $\mathbf{x}$. The second term regularizes the posterior distribution by promoting the distribution of the encoded latent variables $q(\mathbf{z}|\mathbf{x})$ to match the prior $p(\mathbf{z})$. From this point of view, β-VAE (Higgins et al. 2016) scales the regularization term of the ELBO by $\beta > 1$. Since VAE model employs the factorized Gaussian prior $p(\mathbf{z})$, β-VAE encodes more disentangled latent variables by matching the encoded latent variables with the prior $p(\mathbf{z})$ by higher pressure. (Higgins et al. 2016) (Burgess et al. 2018)

$$\begin{aligned}\mathcal{L}_{\beta-\text{VAE}}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z})] \\ -\beta\, D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \mid\mid p(\mathbf{z}))\end{aligned} \tag{5}$$

2.2 JointVAE

VAE and β-VAE employs only continuous latent variables, especially factorized Gaussian, to model the latent variable $\mathbf{z}$. JointVAE (Dupont 2018) generalizes the previous VAE and β-VAE by adopting not only continuous variables but also discrete variables to disentangle the generative factors of the observed data.

Let $\mathbf{z}$ be a continuous latent variable and $\mathbf{c}$ be a discrete latent variable. By assuming conditional independence between the continuous and discrete latent variables, i.e. $q_\phi(\mathbf{z}, \mathbf{c}|\mathbf{x}) = q_\phi(\mathbf{z}|\mathbf{x})\, q_\phi(\mathbf{c}|\mathbf{x})$ and independent prior, i.e. $p(\mathbf{z}, \mathbf{c}) = p(\mathbf{z})\, p(\mathbf{c})$, JointVAE derived an optimization objective (Eq 6) from the β-VAE objective (Eq 5). To prevent the posterior collapse phenomenon of the discrete latent variable, JointVAE applied the capacity control (Burgess

et al. 2018) to the objective.

$$\mathcal{L}_{\text{Joint}}(\theta,\phi) = \mathbb{E}_{q(\mathbf{z}|\mathbf{x},\mathbf{c})}\left[\log p(\mathbf{x} \mid \mathbf{z})\right] - \beta_{\mathbf{z}} \mid D_{KL}(q(\mathbf{z} \mid \mathbf{x}) \mid\mid p(\mathbf{z})) - C_z \mid - \beta_c \mid D_{KL}(q(\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{c})) - C_c \mid \quad (6)$$

JointVAE parametrizes the variational distribution of discrete variable $q_\phi(\mathbf{c}|\mathbf{x})$ by the neural network. Since the sampling process from the categorical distribution is non-differentiable, the reparametrization trick can not be applied directly to the discrete variable. To address this problem, JointVAE employed a differentiable relaxation of discrete variable sampling called Gumbel-Softmax distribution. (Jang, Gu, and Poole 2016) (Maddison, Mnih, and Teh 2016) (Gumbel 1954)

If $\mathbf{c}$ is a discrete or categorical variable with class probabilities $\alpha_1, \alpha_2, \cdots, \alpha_n$, samples from the Gumbel-Softmax distribution can be obtained by sampling $g_k \sim \text{Gumbel}(0,1)$ i.i.d. and transforming the samples to

$$\mathbf{y}_k = \frac{\exp((\log(\alpha_k) + g_k)/\tau)}{\sum_i \exp((\log(\alpha_i) + g_i)/\tau)} \quad (7)$$

where τ denotes a temperature parameter.

3 Discond-VAE

In this section, we describe the motivation and probabilistic formulation of Discond-VAE. Then, we describe how the probabilistic formulation is instantiated by a neural network.

3.1 Motivation

Although JointVAE (Dupont 2018) extends the capability of VAE to encode discrete factors, JointVAE assumes conditional independence of the continuous variable and discrete variable. However, this assumption usually does not hold for the general dataset, e.g., ImageNet and CelebA dataset. Therefore, we generalize the assumption.

Consider a generative modeling problem with the observed data $\mathbf{x}$, the discrete generative factor $\mathbf{c}$, and a set of continuous factors. Some of the continuous factors are independent of the discrete factor $\mathbf{c}$, and the others are not. We refer to the former independent continuous factor as *public generative factor* and the latent variable representing it as a *public variable* $\mathbf{z}$. Likewise, we call the latter dependent continuous factor as *private factor* and the latent variable representing it as a *private variable* $\mathbf{w}$.

For example, consider a toy dataset, which is a subset of the dSprites (Matthey et al. 2017). This toy dataset is a collection of two-dimensional Square and Ellipse images. In the dataset, the images vary in scale and orientation for each shape. Also, the Square images vary in x-position, and the Ellipse images vary in y-position. In this situation, a latent variable that encodes scale and orientation should be independent of the discrete variable which encodes shape. However, the latent variables representing x, y-position should be dependent on the discrete variable. In short, this dataset has the public factor of scale and orientation, and the private factor of x, y-position. We refer to this toy dataset as Cond-Sprites and use this dataset to evaluate Discond-VAE in Sec 4.3.

3.2 Model

Probabilistic Model We propose a modification to JointVAE (Dupont 2018) whose latent variable is composed of the discrete, public, and private variables. Since the public variable represents generative factors shared by all classes and the private variable represents variation within each class, it is natural to assume that the prior $p(\mathbf{z},\mathbf{w},\mathbf{c})$ factorizes to $p(\mathbf{z})$ and $p(\mathbf{w},\mathbf{c})$.

$$p(\mathbf{z},\mathbf{w},\mathbf{c}) = p(\mathbf{z}) \cdot p(\mathbf{w},\mathbf{c}) = p(\mathbf{z}) \cdot p(\mathbf{c}) \cdot p(\mathbf{w} \mid \mathbf{c}) \quad (8)$$

Likewise, the variational distribution $q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})$ is modeled as the following.

$$\begin{aligned} q_\phi(\mathbf{z},\mathbf{w},\mathbf{c} \mid \mathbf{x}) &= q_\phi(\mathbf{z} \mid \mathbf{x}) \cdot q_\phi(\mathbf{w},\mathbf{c} \mid \mathbf{x}) \\ &= q_\phi(\mathbf{z} \mid \mathbf{x}) \cdot q_\phi(\mathbf{c} \mid \mathbf{x}) \cdot q_\phi(\mathbf{w} \mid \mathbf{x},\mathbf{c}) \end{aligned} \quad (9)$$

For our Discond-VAE case, the β-VAE objective (Eq 5) becomes

$$\mathcal{L}_{\text{Cond}}(\theta,\phi) = \mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z},\mathbf{w},\mathbf{c})] - \beta\, D_{KL}(q_\phi(\mathbf{z},\mathbf{w},\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{z},\mathbf{w},\mathbf{c})) \quad (10)$$

The former log-likelihood term stands for the reconstruction error as in the previous VAE models. The latter KL divergence regularizer can be decomposed by the independence assumption.

$$D_{KL}(q_\phi(\mathbf{z},\mathbf{w},\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{z},\mathbf{w},\mathbf{c}) = D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \mid\mid p(\mathbf{z})) + D_{KL}(q_\phi(\mathbf{w},\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{w},\mathbf{c})) \quad (11)$$

Then, we can address the latter KL divergence as the following. (See appendix for proof)

$$\begin{aligned} D_{KL}(q_\phi(\mathbf{w},\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{w},\mathbf{c})) &= \mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[D_{KL}(q_\phi(\mathbf{w} \mid \mathbf{x},\mathbf{c}) \mid\mid p(\mathbf{w} \mid \mathbf{c}))\right] \\ &\quad + D_{KL}(q_\phi(\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{c})) \end{aligned} \quad (12)$$

In brief, the learning objective of Discond-VAE (Eq 10) is expressed as

$$\begin{aligned} \max_{\theta,\phi} \mathcal{L}_{\text{Cond}}(\theta,\phi) &= \mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z},\mathbf{w},\mathbf{c})] \\ &\quad - \beta_{\mathbf{z}} \cdot D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \mid\mid p(\mathbf{z})) \\ &\quad - \beta_{\mathbf{w}} \cdot \mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[D_{KL}(q_\phi(\mathbf{w} \mid \mathbf{x},\mathbf{c}) \mid\mid p(\mathbf{w} \mid \mathbf{c}))\right] \\ &\quad - \beta_{\mathbf{c}} \cdot D_{KL}(q_\phi(\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{c})) \end{aligned} \quad (13)$$

Discond-VAE models the $q_\phi(\mathbf{z}|\mathbf{x}), q_\phi(\mathbf{c}|\mathbf{x})$ by the factorized Gaussian and Gumbel-Softmax as in the JointVAE (Dupont 2018). Moreover, Discond-VAE introduces the Gaussian Mixture encoder to model the joint distribution of the private and discrete variables. Each mode of the Mixture represents the generative factors within a class.

$$p(\mathbf{w} \mid \mathbf{c}) = \prod_i p(\mathbf{w} \mid \mathbf{c} = e_i)^{c_i} \quad (14)$$

$$= \prod_i \mathcal{N}(\mu_i, I)^{c_i} \quad (15)$$

$$q_\phi(\mathbf{w} \mid \mathbf{x},\mathbf{c}) = \prod_i q_\phi(\mathbf{w} \mid \mathbf{x},\mathbf{c} = e_i)^{c_i} \quad (16)$$

$$= \prod_i \mathcal{N}(\mu_i(\mathbf{x},\mathbf{c}), \Sigma_i(\mathbf{x},\mathbf{c}))^{c_i} \quad (17)$$

where $\mathbf{c} = (c_1, c_2, \cdots, c_d) \in \{0,1\}^d$ denote a one-hot sample from the d-dimensional categorical distribution and e_i denote a one-hot vector with ith component is one. Then, the KL divergence term of private variable $\mathbf{w}$ becomes

$$\mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[D_{KL}(q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c}) \mid\mid p(\mathbf{w} \mid \mathbf{c}))\right] \tag{18}$$

$$= \sum_{i=1}^{d} \alpha_i \cdot D_{KL}(q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c} = e_i) \mid\mid p(\mathbf{w} \mid \mathbf{c} = e_i)) \tag{19}$$

where $q_\phi(\mathbf{c}|\mathbf{x}) = (\alpha_1, \alpha_2, \cdots, \alpha_d)$.

Implementation We propose two methods to implement the probabilistic model of Discond-VAE.

First, we can model the posterior distribution $q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c})$ while keeping the discreteness of the categorical variables. For each class $\mathbf{c} = e_i$, the private variable encoder $q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c})$ takes a concatenation of features extracted from the data $\mathbf{x}$ and one-hot encoding of the class e_i to infer the corresponding mode of the Gaussian Mixture.

$$q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c} = e_i) = \mathcal{N}\left(\mu_i(\mathbf{x}, e_i), \Sigma_i(\mathbf{x}, e_i)\right) \tag{20}$$

Note that the private variable encoder infers d-times where d denotes the number of classes. In this case, the exact evaluation of reconstruction error requires generating reconstruction d-times. To ease the computational complexity, we take an approximation of the reconstruction error by generating reconstruction only for the most likely class from $q_\phi(\mathbf{c}|\mathbf{x})$.

$$\begin{aligned}&\mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{w}, \mathbf{c})] \\ &\quad \approx \mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w}|\mathbf{x},\mathbf{c}=e_j)}[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{w}, \mathbf{c} = \boldsymbol{\alpha})]\end{aligned} \tag{21}$$

where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \cdots, \alpha_d)$ denotes the variational distribution of the discrete variable and $j = \arg\max_i q_\phi(\mathbf{c} = e_i|\mathbf{x})$. (Fig 2) We refer to this model by *Discond-VAE-exact*. Note that for the *Discond-VAE-exact* with the perfect classification of confidence 100%, the continuous variable encoder is a combination of the vanila encoder applied to the entire dataset and the class-specific vanila encoder applied only to the corresponding class.

Second, we can approximate the discreteness of the categorical variables by applying the reparametrization trick with the Gumbel-Softmax distribution (Jang, Gu, and Poole 2016) (Maddison, Mnih, and Teh 2016). Specifically, the discrete variable $\mathbf{c}$ of data $\mathbf{x}$ is approximated by a sample $\boldsymbol{\pi} = (\pi_1, \cdots, \pi_d)$ from the Gumbel-Softmax distribution. We refer to this model by *Discond-VAE-approx*.

$$q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c} = e_i) = \mathcal{N}\left(\mu_i(\mathbf{x}, \boldsymbol{\alpha}), \Sigma_i(\mathbf{x}, \boldsymbol{\alpha})\right) \tag{22}$$

$$\begin{aligned}&\mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{w}, \mathbf{c})] \\ &\quad \approx \mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w}|\mathbf{x},\mathbf{c}=\boldsymbol{\pi})}[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{w}, \mathbf{c} = \boldsymbol{\pi})]\end{aligned} \tag{23}$$

For *Discond-VAE-approx model*, we adopt a continuous approximation of the private variable sample from the Gaussian Mixture $q_\phi(\mathbf{w}, \mathbf{c}|\mathbf{x})$ by taking a linear combination of the samples from each mode $q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c} = e_i)$. (Fig 2)

$$\mathbf{w} = \sum_{i}^{d} \pi_i \cdot \mathbf{w}_i \tag{24}$$

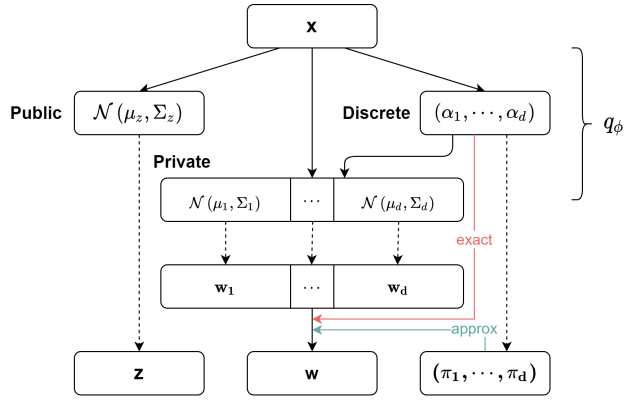


Figure 2: Encoder and Reparametrization trick of Discond-VAE. The dashed lines denote the reparametrization trick from each variational distribution. The Discond-VAE-exact takes the private variable sample from the Mixture mode of the most likely class. The Discond-VAE-approx takes a linear combination of the Gaussian samples from each mode as the private variable sample.

where $\mathbf{w}_i \sim q_\phi(\mathbf{w}|\mathbf{x}, \mathbf{c} = e_i)$.

Both types of Discond-VAE apply a reparametrization trick to the discrete variable $q_\phi(\mathbf{c}|\mathbf{x})$ in estimating the KL divergence of private variable (Eq 18). Also, two Discond-VAE implementations are optimized with the capacity objective (Burgess et al. 2018) as in JointVAE to prevent a discrete variable from posterior collapsing. Hence, the learning objective of Discond-VAE becomes

$$\begin{aligned}&\max_{\theta,\phi} \mathcal{L}_{\text{Cond}}(\theta, \phi) = \mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{w}, \mathbf{c})] \\ &- \beta_{\mathbf{z}} \left|D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \mid\mid p(\mathbf{z})) - C_{\mathbf{z}}\right| \\ &- \beta_{\mathbf{w}} \left|\mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[D_{KL}(q_\phi(\mathbf{w} \mid \mathbf{x}, \mathbf{c}) \mid\mid p(\mathbf{w} \mid \mathbf{c}))\right] - C_{\mathbf{w}}\right| \\ &- \beta_{\mathbf{c}} \left|D_{KL}(q_\phi(\mathbf{c} \mid \mathbf{x}) \mid\mid p(\mathbf{c})) - C_{\mathbf{c}}\right|\end{aligned} \tag{25}$$

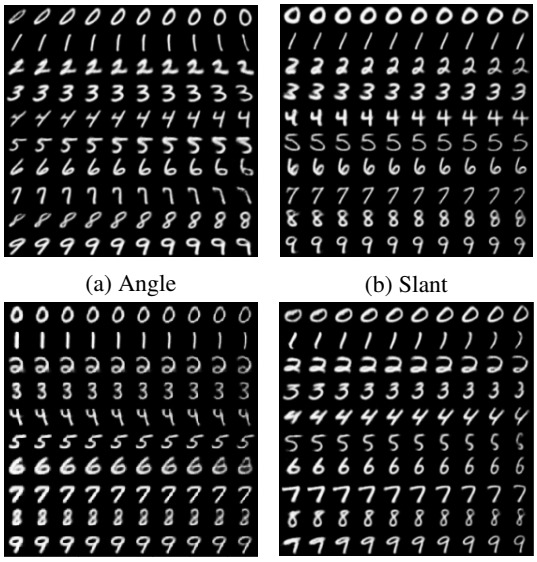
The means of each mode in Mixture prior μ_i are hyperparameters determined at the initialization. The Discond-VAE-approx model sets the μ_i by the random samples from the standard Gaussian $\mathcal{N}(0, 1)$. Interestingly, the Discond-VAE-exact models show similar performance for the $\mu_i = 0$ with a smaller variance. Therefore, we set the $\mu_i = 0$ for the Discond-VAE-exact models in Sec 4.

4 Experiments

We evaluate the Discond-VAE model on the MNIST, dSprites (Matthey et al. 2017), and CondSprites datasets. For each dataset, we assess the Discond-VAE model in an unsupervised manner. For each quantitative scores, we evaluate the model ten times randomly, and report the means, standard deviations, and the best scores. (See appendix for the full architecture and training hyperparameters)

4.1 MNIST

Dataset The MNIST dataset consists of 28×28 handwritten digit images. For the MNIST dataset, the digit-type of



(a) Angle (b) Slant

(c) Thickness (d) Width

Figure 3: Public variable traversal on MNIST

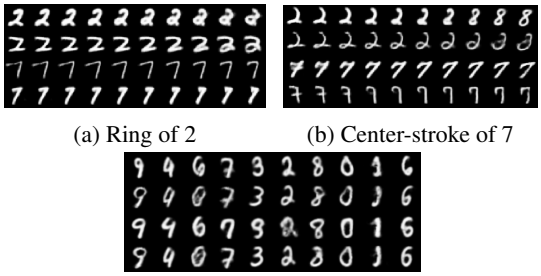
each image corresponds to a discrete generative factor.

Latent Traversal We observed the latent traversals of the Discond-VAE to evaluate disentanglement property qualitatively. For the continuous latent variables, each row corresponds to the latent traversals of an axis over a given example. For the discrete variable, each row shows the one-hot traversal of the discrete variable.

The Discond-VAE shows a smooth variation in angle, slant, thickness, and width of the reconstruction images as we traverse the public variable in Fig 3. In discrete variable traversal (Fig 4c), we can observe a transition in digit-type of given examples. These results demonstrate that the Discond-VAE can disentangle the public and discrete generative factors from the MNIST dataset. Moreover, the Discond-VAE discovers the class-specific variation of the digit-type two and seven. The Fig 4a and 4b shows the private variable traversal of the Discond-VAE. Each private variable in Fig 4a and 4b represents the ring of digit-type 2 and the center-stroke of digit-type 7, respectively. Since these two variations are exclusive to each class, the latent traversals of these two variables show relatively minor or irrelevant variations to the other class images.

Accuracy The MNIST dataset has a discrete generative factor of digit-type. Thus, we assess whether the Discond-VAE can disentangle the digit-type from continuous factors into the discrete variable. We consider the discrete variable encoder $q(\mathbf{c} \mid \mathbf{x})$ as an unsupervised majority vote classifier and evaluate its accuracy.

Table 1 shows the unsupervised classification accuracy of each model. Since the continuous latent variable of the JointVAE is equivalent to the public variable of Discond-VAE, we compare the models with the same public dimension. For each Discond-VAE model, we introduced a three-dimensional private latent variable. Both types of Discond-VAE shows at least comparable accuracy to the JointVAE while representing the additional private variables. Considering that the models with smaller continuous variables show higher accuracy in Table 1, the Discond-VAE model with the same continuous dimension might outperform the JointVAE further. Unlike the JointVAE, the Discond-VAE can disentangle the class-dependent continuous generative factors. This capability results in the more disentangled representation of the discrete variable, which leads to higher accuracy.



(a) Ring of 2 (b) Center-stroke of 7

(c) Discrete variable traversal

Figure 4: Private and Discrete variable traversal on MNIST

Method	Pb	Mean (std)	Best
JointVAE	10	0.686 (0.092)	**0.809**
	4	0.708 (0.059)	0.792
Discond-VAE-exact	10	0.686 (0.078)	0.807
	4	0.722 (0.099)	0.876
Discond-VAE-approx	10	**0.739** (0.046)	0.793
	4	**0.772** (0.076)	**0.879**

Table 1: Unsupervised classification accuracy for MNIST. Pb represents the dimension of the continuous variable for the JointVAE, and the public variable dimension for the Discond-VAE.

Method	Pb	Mean (std)	Best
JointVAE	6	0.448*(0.039)	0.531*
	4	0.440*(0.039)	**0.541***
Discond-VAE-exact	6	0.389(0.040)	0.444
	4	0.369(0.010)	0.381
Discond-VAE-approx	6	**0.456** (0.060)	**0.559**
	4	**0.478** (0.031)	0.533

Table 2: Unsupervised classification accuracy for dSprites. * indicates the results from (Jeong and Song 2019).

4.2 dSprites

Dataset The dSprites (Matthey et al. 2017) is a synthetic dataset to evaluate the disentanglement property of a model. Each sample of the dataset is a two-dimensional shape image generated from five generative factors. The dSprites dataset has one discrete generative factor of shape (square, ellipse, heart), and four continuous generative factors of scale, orientation, and position in x, y axis.

Generative factors	Shape	
	Square	Ellipse
Scale	O	O
Orientation	O	O
Position X	O	X
Position Y	X	O

Table 3: Generative factors of the CondSprites.

Quantitative Result As a quantitative evaluation, we assess the Discond-VAE trained on the dSprites by the accuracy and two disentanglement metrics (FactorVAE metric (Kim and Mnih 2018) and MIG (Chen et al. 2018)). As in the MNIST, we evaluate the discrete variable encoder as a majority vote classifier according to the shape. Most of the disentanglement metrics assume that the latent variables are continuous. For example, the β-VAE metric (Higgins et al. 2016) and FactorVAE metric (Kim and Mnih 2018) measures the degree of disentanglement from the accuracy of the classifier predicting the generative factor based on the variance of each axis of representation. However, our Discond-VAE model and JointVAE adopts the discrete variable to represent the discrete generative factor. Therefore, we evaluated the disentanglement metric on the continuous latent variable based on the continuous generative factors.

The dSprites is a synthetic dataset designed to have a five *independent* generative factors. Hence, the JointVAE is a more appropriate generative model to represent the dSprites compared to the Discond-VAE. Nevertheless, the Discond-VAE-approx models show a comparable classification accuracy in Table 2, and a similar disentanglement metric of the continuous variables in Table 5. Although the Discond-VAE-exact models show a relatively lower classification accuracy, the Discond-VAE-exact model with the six-dimensional public variable shows a similar degree of disentanglement.

4.3 CondSprites

Dataset To evaluate disentangling ability further, we constructed a CondSprites dataset from the dSprites dataset (Matthey et al. 2017). The CondSprites dataset is designed to mimic the class-independent generative factor and intra-class variation generative factor of the real-world data. (See Sec 3.1 and Table 3 for details) The CondSprites has 15,360 two-dimensional images consisting of 7,680 for Square and Ellipse, respectively.

Quantitative Result As in the dSprites dataset, we evaluate the Discond-VAE by the accuracy and disentanglement metrics. Since the dSprites dataset has independent continuous generative factors, we evaluated each model by the disentanglement scores of continuous variables. However, the CondSprites dataset has class-dependent latent variables. Therefore, we propose a conditional disentanglement evaluation. We define the conditional disentanglement metric as an expectation over discrete variables of the classwise disentanglement metrics. By the conditional disentanglement metric, we can assess the disentanglement of private generative factors as well as the public factors.

$$\text{Conditional Metric} = \mathbb{E}_{p(\mathbf{c})}\left[\text{Metric}(\mathbf{X_c})\right] \tag{26}$$

where $\mathbf{X_c}$ denotes the examples from $\mathbf{X}$ with the class $\mathbf{c}$.

Method	Pub	Mean (std)	Best
JointVAE	10	0.617(0.068)	0.720
	5	0.599(0.064)	0.704
Discond-VAE-exact	10	0.630(0.06)	0.763
	5	0.648(0.083)	0.805
Discond-VAE-approx	10	**0.684**(0.192)	**1.00**
	5	**0.734**(0.151)	**0.951**

Table 4: Unsupervised classification accuracy for CondSprites. Discond-VAE models outperform the JointVAE for each public variable dimension.

We report the classification accuracy on the CondSprites in Table 4. Compared to the dSprites where the JointVAE and Discond-VAE showed similar performance, the Discond-VAE outperforms the JointVAE by a significant margin on the CondSprites. In particular, the Discond-VAE-approx model with a five-dimensional public variable outperforms the JointVAE with the same public variable dimension by 13% in mean, and 25% in the best scores. Furthermore, the Discond-VAE-exact model achieves much higher disentanglement scores in both metrics than the JointVAE on the CondSprites in Table 5. The Discond-VAE-approx model shows the higher FactorVAE metric and comparable MIG to the JointVAE. The disentanglement results demonstrate that the Discond-VAE can disentangle the private variables. Moreover, by disentangling the private variables from the discrete variable, the Discond-VAE can attain a more disentangled discrete representation, which is proved by a higher classification accuracy in Table 4.

5 Related Works

Extracting disentangled features from data without supervision is an important task for representation learning. (Bengio, Courville, and Vincent 2013) Several VAE variants adopting continuous latent variables are proposed to obtain more disentangled representations. For example, β-VAE (Higgins et al. 2016) increases a disentangling pressure of VAE by increasing the weight of the KL divergence between the variational posterior $q(\mathbf{z}|\mathbf{x})$ and the prior $p(\mathbf{z})$.

The KL divergence regularizer of β-VAE penalizes not only the total correlation (TC) of the aggregate posterior $q(\mathbf{z})$, which induces the factorized posterior, but also the mutual information between the data and the latent variables. Since penalizing the mutual information is detrimental to extract meaningful features, several works proposed penalizing TC only in various ways. FactorVAE (Kim and Mnih 2018) penalizes TC by introducing an auxiliary discriminator which approximates TC. β-TCVAE (Chen et al. 2018) adopts a mini-batch weighted sampling method to estimate TC. DIP-VAE (Kumar, Sattigeri, and Balakrishnan 2017) suggests an alternative to match the marginal latent distribution $q(\mathbf{z})$ to the prior $p(\mathbf{z})$. The alternative is matching the covariance of the two distributions.

Method	Pb	dSprites				Pb	CondSprites			
		FactorVAE		MIG			Cond-FactorVAE		Cond-MIG	
		Mean (std)	Best	Mean (std)	Best		Mean (std)	Best	Mean (std)	Best
JointVAE	6	0.921(0.002)	0.925	**0.336**(0.002)	0.337	10	0.743 (0.121)	0.870	0.188 (0.075)	0.284
	4	**0.989**(0.004)	**0.991**	0.223(0.022)	0.241	5	0.734 (0.041)	0.761	0.243 (0.041)	0.305
Disc-exact	6	0.916(0.000)	0.916	0.309(0.024)	0.338	10	**0.96** (0.059)	**1.00**	**0.291** (0.065)	**0.322**
	4	0.832(0.037)	0.888	**0.355**(0.017)	**0.382**	5	**0.985**(0.024)	**1.00**	**0.385** (0.124)	**0.466**
Disc-approx	6	**0.922** (0.065)	**0.998**	0.290(0.080)	**0.391**	10	0.948 (0.036)	0.992	0.208 (0.059)	0.315
	4	0.934(0.050)	0.9876	0.225(0.075)	0.379	5	0.913 (0.051)	0.990	0.183 (0.059)	0.302

Table 5: Disentanglement metrics on dSprites and CondSprites. (The higher, the better) The best scores for each combination of a dataset, Pb, and metric are shown in boldface type. On the dSprites with independent generative factors, JointVAE and Discond-VAE show similar scores except for the Discond-VAE-exact with the four-dimensional public variable. On the CondSprites, which has class-dependent and class-independent factors, the Discon-VAE outperforms the JointVAE.

In addition, Bayes-Factor-VAE (Kim et al. 2019) claims that the relevant variable and nuisance variable require a different prior distribution. Bayes-Factor-VAE promotes disentangled features by introducing hyper-priors on the variances of Gaussian prior. HFVAE (Esmaeili et al. 2019) used a two-level hierarchical objective to control the independence between groups of variables and independence between each variable in the same group.

Recently, a number of works to model intrinsic discreteness of the real-world data are proposed. Some of these works proposed representing the discrete variable by modeling the continuous variable as a multimodal distribution or tree-structured model. (Dilokthanakul et al. 2016) represents continuous variables as a Gaussian Mixture and infers the discrete variable by Bayes rule. CascadeVAE (Jeong and Song 2019) proposed an iterative optimization method to minimize the TC of continuous variables and alternating optimization method between discrete and continuous variables to train the model. CascadeVAE infers the discrete variable via the inner maximization step over the discrete variables. Moreover, (Goyal et al. 2017) and LTVAE (Li et al. 2018) encode the latent variable as a tree-structured model and learns the tree structure from the data themselves. (Goyal et al. 2017) employs nested Chinese Restaurant Process (Blei, Griffiths, and Jordan 2010) to accommodate a hierarchical prior to the data. LTVAE adjusts a tree-structure of the latent variable via EM algorithm.

In comparison, JointVAE (Dupont 2018) and InfoCatVAE (Pineau and Lelarge 2018) have an explicit encoder to encode discrete variable. JointVAE (Dupont 2018) proposed a method of jointly training continuous and discrete variables. JointVAE integrated non-differentiable sampling of the discrete variable into the gradient descent framework by applying Gumbel-Softmax. (Jang, Gu, and Poole 2016) (Maddison, Mnih, and Teh 2016) However, JointVAE has a limitation of assuming that the discrete and continuous variables are independent of each other. By introducing the private latent variables, our Discond-VAE can represent the dependent discrete and continuous variables. InfoCatVAE (Pineau and Lelarge 2018) encodes a discrete variable and uses it to encode a conditional distribution of continuous variable $q(\mathbf{w}|\mathbf{x}, \mathbf{c})$. In this respect, the private latent variable of our proposed Discond-VAE and InfoCatVAE have a similar probabilistic formulation. However, InfoCatVAE employs Gaussian distribution as a class-conditioned continuous variable, and regularizes the variational distribution with a Mixture of Gaussian. The axes of the Gaussian distribution are divided into groups, and each group is intended to encode the intra-class variation of a particular class as the private variable does. However, since the entire Gaussian distribution is fed to the decoder, each group is obliged to represent a certain latent variable even for the irrelevant classes. On the other hand, our proposed Discond-VAE adopts the Mixture of Gaussian to the variational distribution and the prior. Since Discond-VAE separates each class-conditioned continuous variable on each mode, the private variable only represents a variation within a class. Therefore, Discond-VAE can promote more disentangled representation compared to InfoCatVAE, even only with the private variable.

6 Conclusion

We proposed Discond-VAE for learning the public and private continuous generative factors and the discrete generative factor from the data. We developed a probabilistic framework and the learning objective for the Discond-VAE, and suggested two implementations of the framework according to how the discrete variable is addressed. Also, we proposed the CondSprites dataset to evaluate the disentanglement capacity for the class-dependent generative factors of a model. Then, we evaluated both types of the Discond-VAE model on the MNIST, dSprites, and CondSprites. The experiment results prove that the Discond-VAE model can disentangle the class-dependent and class-independent factors in an unsupervised manner. Moreover, the Discond-VAE shows a similar degree of disentanglement compared to the previous model even on the dSprites with only independent generative factors.

If we consider the public variable as a private variable over the discrete variable of one class, the Discond-VAE has a natural hierarchical extension. It would be interesting to develop the hierarchical extension of the Discond-VAE in future work.

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A Formulation

A.1 ELBO Derivation

$$\log p(\mathbf{x}) = \tag{27}$$

$$\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\left[\log\left(\frac{p(\mathbf{x},\mathbf{z})}{q(\mathbf{z}\mid\mathbf{x})}\right)\right] + D_{KL}(q(\mathbf{z}\mid\mathbf{x})\,||\,p(\mathbf{z}\mid\mathbf{x})) \tag{28}$$

$$\geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\left[\log\left(\frac{p(\mathbf{x},\mathbf{z})}{q(\mathbf{z}\mid\mathbf{x})}\right)\right] \tag{29}$$

$$= \mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\left[\log p(\mathbf{x}\mid\mathbf{z})\right] - D_{KL}(q(\mathbf{z}\mid\mathbf{x})\,||\,p(\mathbf{z})) \tag{30}$$

$$= \mathcal{L}_{\text{VAE}}(\theta,\phi) \tag{31}$$

A.2 Private Variable KL divergence

$$\begin{aligned}
&D_{KL}(q_\phi(\mathbf{w},\mathbf{c}\mid\mathbf{x})\,||\,p(\mathbf{w},\mathbf{c})) = \\
&= \mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\mathbb{E}_{q_\phi(\mathbf{w}|\mathbf{x},\mathbf{c})}[\log\frac{q_\phi(\mathbf{w}\mid\mathbf{x},\mathbf{c})\cdot q_\phi(\mathbf{c}\mid\mathbf{x})}{p(\mathbf{w}\mid\mathbf{c})\cdot p(\mathbf{c})}] \\
&= \mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[\mathbb{E}_{q_\phi(\mathbf{w}|\mathbf{x},\mathbf{c})}[\log\frac{q_\phi(\mathbf{w}\mid\mathbf{x},\mathbf{c})}{p(\mathbf{w}\mid\mathbf{c})}] + \log\frac{q_\phi(\mathbf{c}\mid\mathbf{x})}{p(\mathbf{c})}\right] \\
&= \mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[\mathbb{E}_{q_\phi(\mathbf{w}|\mathbf{x},\mathbf{c})}[\log\frac{q_\phi(\mathbf{w}\mid\mathbf{x},\mathbf{c})}{p(\mathbf{w}\mid\mathbf{c})}]\right] \\
&\qquad + D_{KL}(q_\phi(\mathbf{c}\mid\mathbf{x})\,||\,p(\mathbf{c}))
\end{aligned} \tag{32}$$

B Implementation details

B.1 Network Architecture

We use similar Network architectures in JointVAE (Dupont 2018). The only modification is that the Discond-VAE-exact model embeds the public variable and the private and discrete variables separately. (Table 6)

B.2 Training details

As a reminder, the learning objective of Discond-VAE is expressed as the following.

$$\begin{aligned}
&\max_{\theta,\phi}\mathcal{L}_{\text{Cond}}(\theta,\phi) = \mathbb{E}_{q_\phi(\mathbf{z},\mathbf{w},\mathbf{c}|\mathbf{x})}[\log p_\theta(\mathbf{x}\mid\mathbf{z},\mathbf{w},\mathbf{c})] \\
&- \beta_{\mathbf{z}}\left|D_{KL}(q_\phi(\mathbf{z}\mid\mathbf{x})\,||\,p(\mathbf{z})) - C_{\mathbf{z}}\right| \\
&- \beta_{\mathbf{w}}\left|\mathbb{E}_{q_\phi(\mathbf{c}|\mathbf{x})}\left[D_{KL}(q_\phi(\mathbf{w}\mid\mathbf{x},\mathbf{c})\,||\,p(\mathbf{w}\mid\mathbf{c}))\right] - C_{\mathbf{w}}\right| \\
&- \beta_{\mathbf{c}}\left|D_{KL}(q_\phi(\mathbf{c}\mid\mathbf{x})\,||\,p(\mathbf{c})) - C_{\mathbf{c}}\right|
\end{aligned} \tag{33}$$

For both Discond-VAE models, we apply the linear scheduling of capacity (Burgess et al. 2018) as in the JointVAE. Each capacity $C_{\mathbf{z}}, C_{\mathbf{w}}, C_{\mathbf{c}}$ is linearly increased from 0 to C in the iteration hyperparameters. For the JointVAE, we applied the same hyperparameters as in (Dupont 2018) for the MNIST and dSprites. All models in the paper are optimized by Adam optimizer. The parameters for the Adam optimizer are betas=(0.9, 0.999), eps=1e-08 with no weight decay, which is the default setting in the PyTorch library.

B.3 Discond-VAE-exact Hyperparameters

MNIST

- Epochs: 100
- Batch size: 64
- Optimizer: Adam with learning rate 5e-4

dSprites

- Epochs: 30
- Batch size: 64
- Optimizer: Adam with learning rate 5e-4

CondSprites

- Epochs: 200
- Batch size: 64
- Optimizer: Adam with learning rate 5e-4

All other hyperparameters in Table 8.

Hyperparameters Search range

- Learning rate - { 5e-4}
- $(\beta_{\mathbf{z}} = \beta_{\mathbf{w}})$ - {5, 10,15,20,25 30, 50,100,200}
- $\beta_{\mathbf{c}}$ - {1,3,5,10,20,30,50,100,200 }
- $(C_{\mathbf{z}} = C_{\mathbf{w}})$ - { 5, 10, 20,30,50 }
- $C_{\mathbf{c}}$ - { 1,1.1 5, 10,25,50 }

B.4 Discond-VAE-approx Hyperparameters

MNIST

- Epochs: 100
- Batch size: 64
- Optimizer: Adam with learning rate 2e-3

dSprites

- Epochs: 20
- Batch size: 64
- Optimizer: Adam with learning rate 1e-3

CondSprites

- Epochs: 300
- Batch size: 64
- Optimizer: Adam with learning rate 1e-3

All other hyperparameters in Table 9.

Hyperparameters Search range

- Learning rate - { 1e-3, 2e-3 }
- $\beta_{\mathbf{z}}$ - {10, 20, 30, 50}
- $\beta_{\mathbf{z}} : (\beta_{\mathbf{w}} = \beta_{\mathbf{c}})$ - { 2:1, 1:1, 1:2 }
- $(C_{\mathbf{z}} = C_{\mathbf{w}})$ - { 5, 10, 20 }
- $C_{\mathbf{c}}$ - { 1, 5, 10 }

Encoder			Decoder	
	1×32 Conv 4×4, stride 2		**Public**	**Private**
	(32×32 Conv 4×4, stride 2)		Linear Pb × 128	Linear (d * Pr + d) × 128
	32×64 Conv 4×4, stride 2		Linear 256 × (64 * 4 * 4)	
	64×64 Conv 4×4, stride 2		(64 × 64 Conv Transpose 4×4, stride 2)	
	Linear (64 * 4 * 4) × 256		64 × 32 Conv Transpose 4×4, stride 2	
Public	**Private**	**Discrete**	32 × 32 Conv Transpose 4×4, stride 2	
Linear 256 × Pb	Linear (256+d) × d * Pr	Linear 256 × d	32 × 1 Conv Transpose 4×4, stride 2	

Table 6: Discond-VAE-exact architecture. For each layers, $a \times b$ at the front represents that the layer has a in-channels and b out-channels. The layers in () are added for the dSprites and CondSprites. d denotes the dimension of the discrete variable. Pr and Pb represent the dimension of the private and public variables, respectively.

Encoder			Decoder
	1×32 Conv 4×4, stride 2		Linear (Pb + Pr + d) ×256
	(32×32 Conv 4×4, stride 2)		Linear 256 × (64 * 4 * 4)
	32×64 Conv 4×4, stride 2		(64 × 64 Conv Transpose 4×4, stride 2)
	64×64 Conv 4×4, stride 2		64 × 32 Conv Transpose 4×4, stride 2
	Linear (64 * 4 * 4) × 256		32 × 32 Conv Transpose 4×4, stride 2
Public	**Private**	**Discrete**	32 × 1 Conv Transpose 4×4, stride 2
Linear 256 × Pb	Linear (256+d) × d * Pr	Linear 256 × d	

Table 7: Discond-VAE-approx architecture. The same notation as in Table 6 is applied.

	MNIST10	MNIST4	dSprites6	dSprites4	CondSprites10	CondSprites5
Pb	10	4	6	4	10	5
Pr	3	3	2	2	3	3
d	10	10	3	3	2	2
$\beta_{\mathbf{z}}$	25	25	200	100	30	30
$\beta_{\mathbf{w}}$	25	25	200	100	30	30
$\beta_{\mathbf{c}}$	5	5	200	100	30	30
$C_{\mathbf{z}}$	5	5	20	20	30	30
$C_{\mathbf{w}}$	5	5	20	20	30	30
$C_{\mathbf{c}}$	25	25	1.1	1.1	5	5
iteration	25000	25000	300000	300000	25000	25000

Table 8: Hyperparameters of DiscondVAE-Exact. Capacity increased linearly from 0 to C in an iteration.

	MNIST10	MNIST4	dSprites6	dSprites4	CondSprites10	CondSprites5
Pb	10	4	6	4	10	5
Pr	3	3	2	2	3	3
d	10	10	3	3	2	2
$\beta_{\mathbf{z}}$	30	20	20	20	10	10
$\beta_{\mathbf{w}}$	60	40	40	40	20	20
$\beta_{\mathbf{c}}$	60	40	40	40	20	20
$C_{\mathbf{z}}$	10	10	10	10	20	20
$C_{\mathbf{w}}$	10	10	10	10	20	20
$C_{\mathbf{c}}$	10	5	5	5	5	5
iteration	25000	25000	300000	300000	25000	25000

Table 9: Hyperparameters of DiscondVAE-approx. Capacity increased linearly from 0 to C in an iteration.

	MNIST10	MNIST5	dSprites6	dSprites4	CondSprites10	CondSprites5
Pb	10	4	6	4	10	5
d	10	10	3	3	2	2
$\beta_{\mathbf{z}}$	30	30	150	150	30	30
$\beta_{\mathbf{c}}$	30	30	150	150	30	30
$C_{\mathbf{z}}$	5	5	40	40	30	30
$C_{\mathbf{c}}$	5	5	1.1	1.1	5	5
iteration	25000	25000	300000	300000	300000	300000

Table 10: Hyperparameters of JointVAE. Capacity increased linearly from 0 to C in an iteration.

Generative factors	Shape	
	Square	Ellipse
Scale	O	O
Orientation	O	O
Position X	O	X
Position Y	X	O

Table 11: Generative factors of the CondSprites.

B.5 JointVAE Hyperparameters

MNIST

- Epochs: 100
- Batch size: 64
- Optimizer: Adam with learning rate 5e-4

dSprites

- Epochs: 30
- Batch size: 64
- Optimizer: Adam with learning rate 5e-4

CondSprites

- Epochs: 200
- Batch size: 64
- Optimizer: Adam with learning rate 5e-4

All other hyperparameters in Table 10.

B.6 Experiments environment

- CPU: Intel(R) Xeon(R) CPU E5-2620 v4 @ 2.10GHz
- GPU: GeForce GTX 1080
- GPU Memory Usage: Around 600 $\sim$ 700 MB depending on datasets.
- OS: Ubuntu 16.04.3 LTS
- Important relevant libraries:
  - JointVAE implementation : Implementation by author (Dupont 2018)
  - Disentanglement metric : Implemented based on Pytorch Implementation of disentanglement_lib (Locatello et al. 2019) - (Abdi, Abolmaesumi, and Fels 2019)
  - PyTorch (Paszke et al. 2017) $>=$ 1.0

C CondSprites Details

The CondSprites is a subset of the dSprites (Matthey et al. 2017) designed to model the dependence between the continuous and discrete generative factors. We removed the Heart shape to maintain a reasonable amount of examples. Since the Square images do not vary in the y-position in Table 3, we fix the Square images on the center of y-axis. In other words, the generative factor for the y-position of Square images in the CondSprites is fixed to 16 (mean of the range(0, 32). Likewise, the generative factor for the x-position of every Ellipse image is fixed to 16. The total number of CondSprites examples is $15360 = 6(\text{Scale}) * 40(\text{Orientation}) * (32(x \text{ for Squares}) + 32(y \text{ for Ellipses}))$.